Cooling the City

Cities that plan for heat are rethinking streets, roofs and parks at the same time. A street lined with trees can be several degrees cooler at noon than one of bare asphalt, and the difference grows through the evening.

The first step is usually a map. Satellite images show which blocks absorb the most sunlight, and those maps often match the neighborhoods with the fewest trees and the oldest housing.

Painting roofs white is cheap and fast. It lowers the temperature inside top-floor flats, which matters most for older residents who cannot easily leave a hot building during a heat wave.

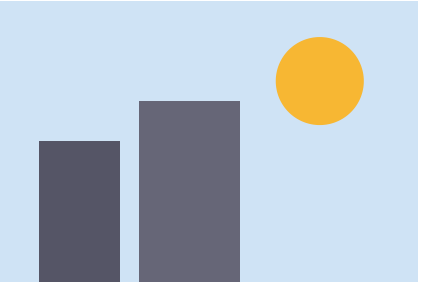

Figure 1. Surface temperature on a July afternoon.

What works

Parks do more than shade. Grass and soil release moisture as they warm, and a large park can cool the streets around it for several hundred meters, especially at night.

Water helps too. Fountains, splash pads and restored streams give people a place to cool down, although they need careful design to avoid wasting water in dry seasons.

None of these measures works alone. The cities with the best results combine them, and they measure the effect street by street so the budget goes where it helps most.